

Support Vector Machine

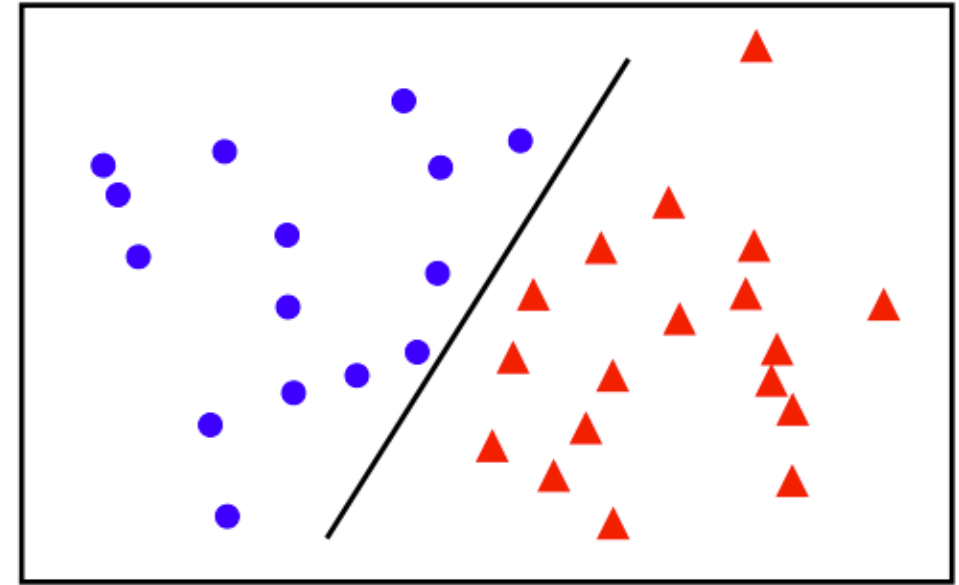
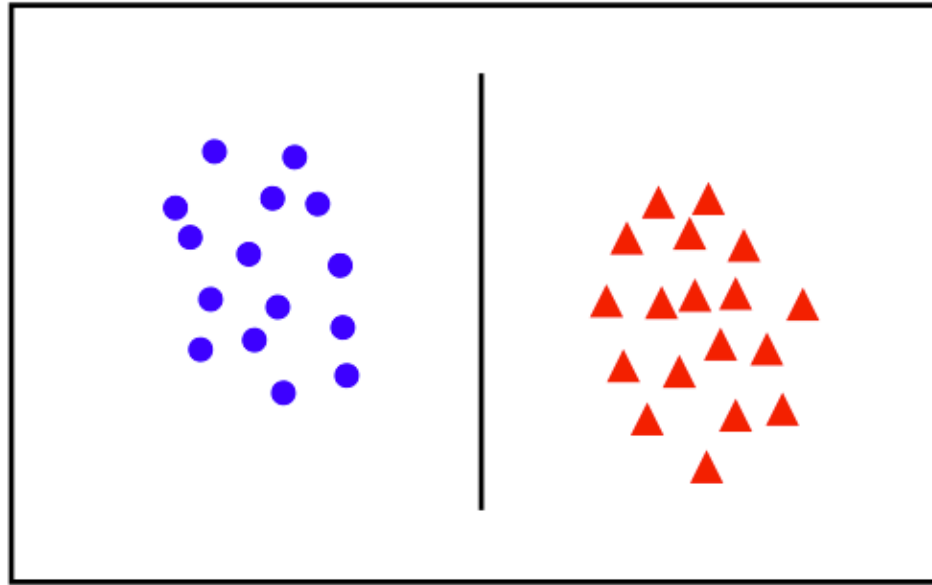
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Georgia Tech

Outline

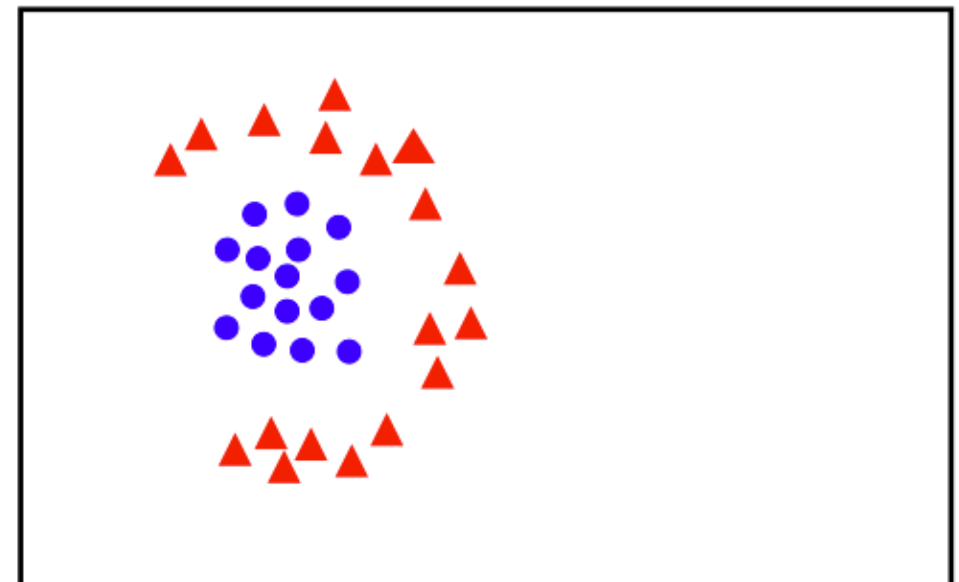
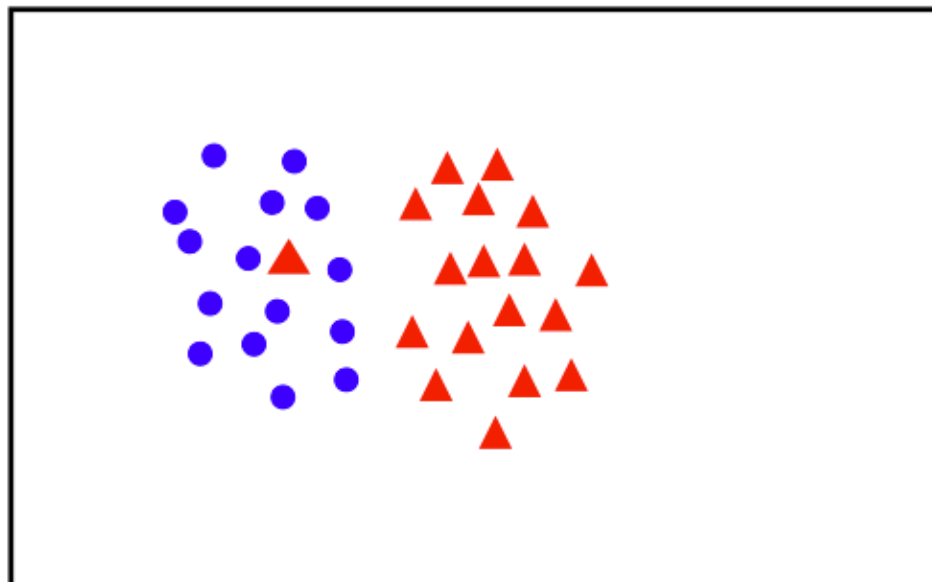
- Precursor: Linear Classifier and Perceptron ←
- Support Vector Machine
- Parameter Learning

Linear Separability

linearly
separable



not
linearly
separable

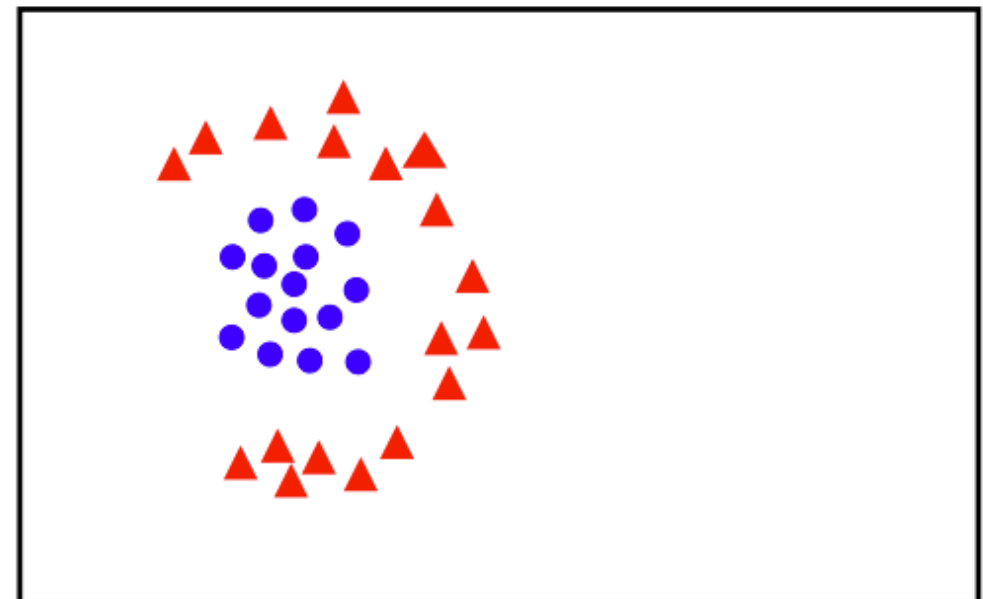
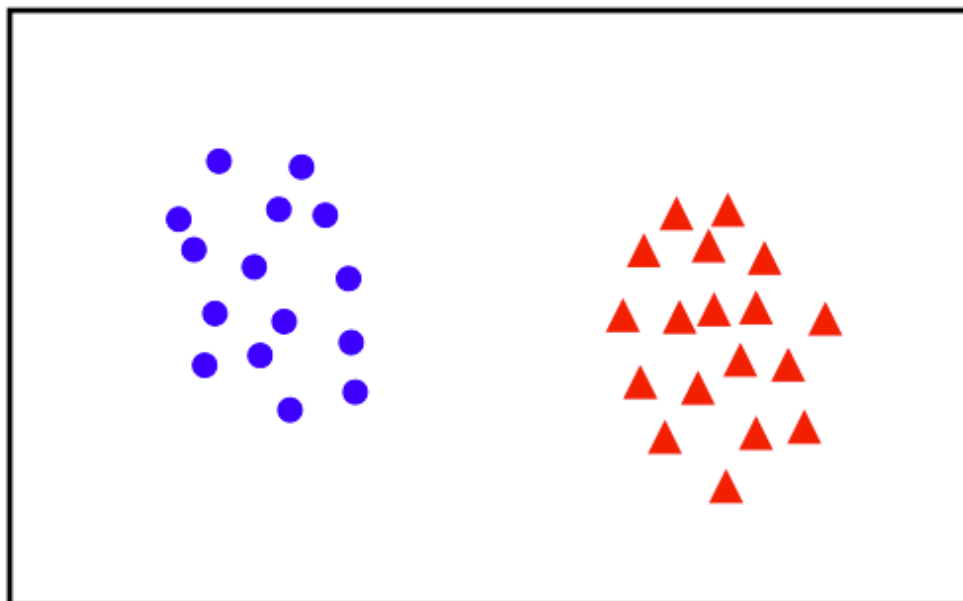


Binary Classification

Given training data (x^{i}, y^{i}) for $i = 1 \dots N$, with $x^{i} \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

$$f(x^{i}) \begin{cases} \geq 0 & +1 \\ < 0 & -1 \end{cases}$$

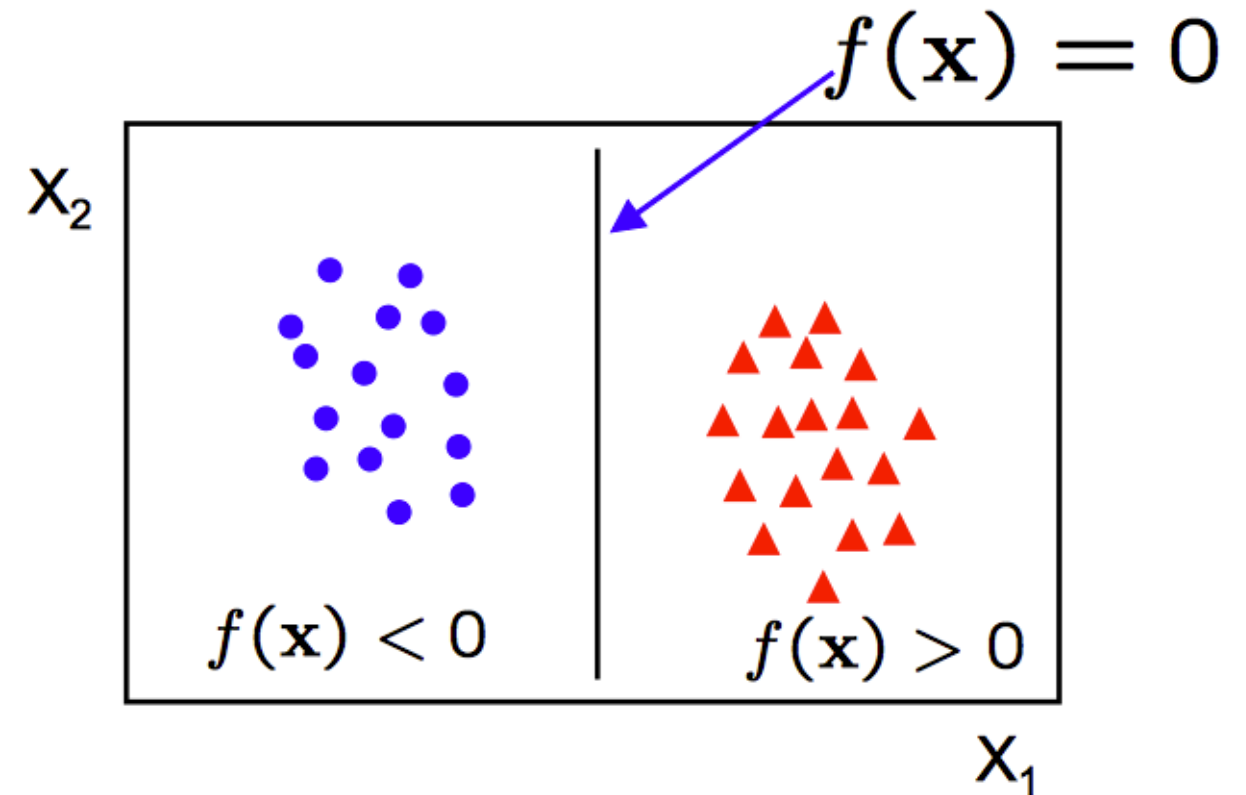
i.e. $y^{i} f(x^{i}) > 0$ for a correct classification.



Linear Classifier

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$

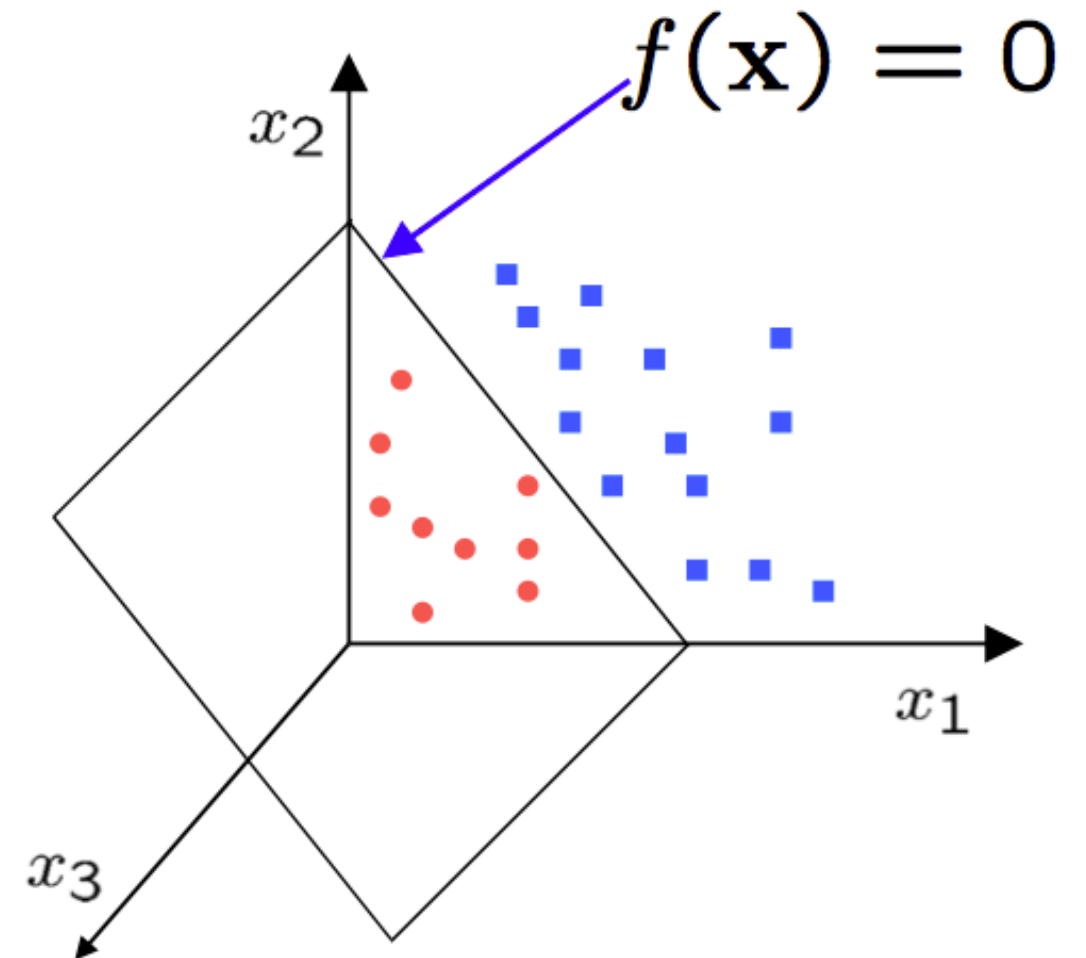


- in 2D the discriminant is a line
- θ is the **normal** to the line, and θ_0 the **bias**
- θ is known as the **weight vector**

Linear Classifier (higher dimension)

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$



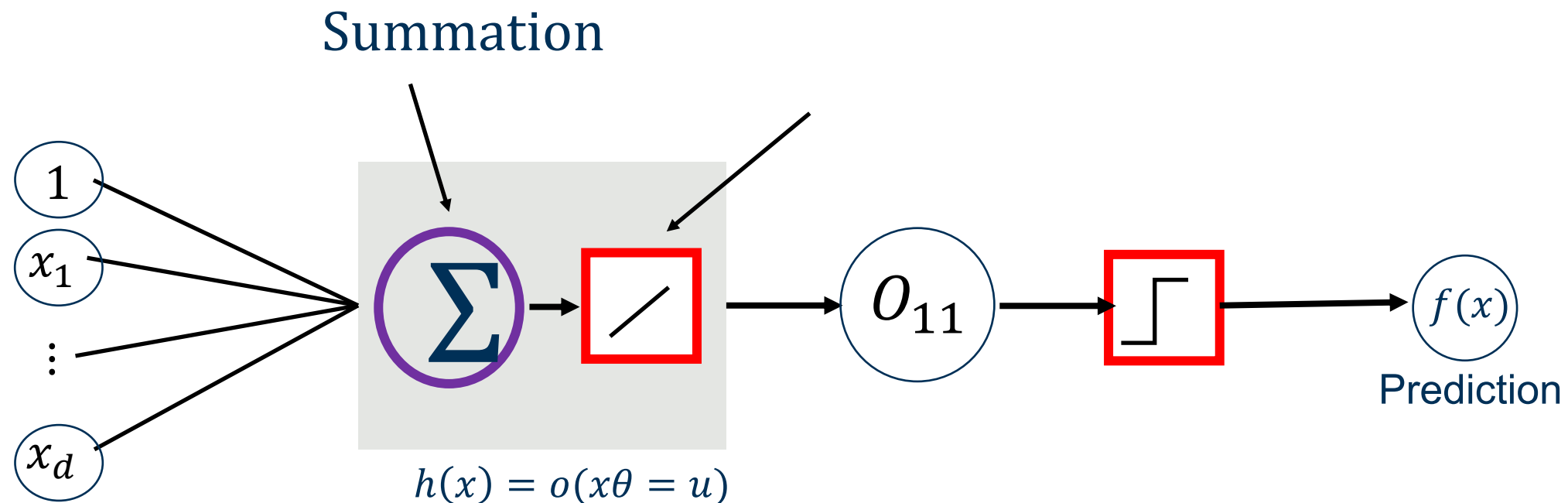
- in 3D the discriminant is a plane, and in nD it is a hyperplane

The Perceptron Classifier

Considering $\mathbf{x}$ is linearly separable and $\mathbf{y}$ has two labels of $\{-1, 1\}$

$$f(x^{\{i\}}) = x^{\{i\}} \theta \quad \text{Bias is inside } \theta \text{ now}$$

How can we separate datapoints with label 1 from datapoints with label -1 using a line?



The Perceptron Classifier

Perceptron Algorithm:

- Initialize θ
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
- If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified

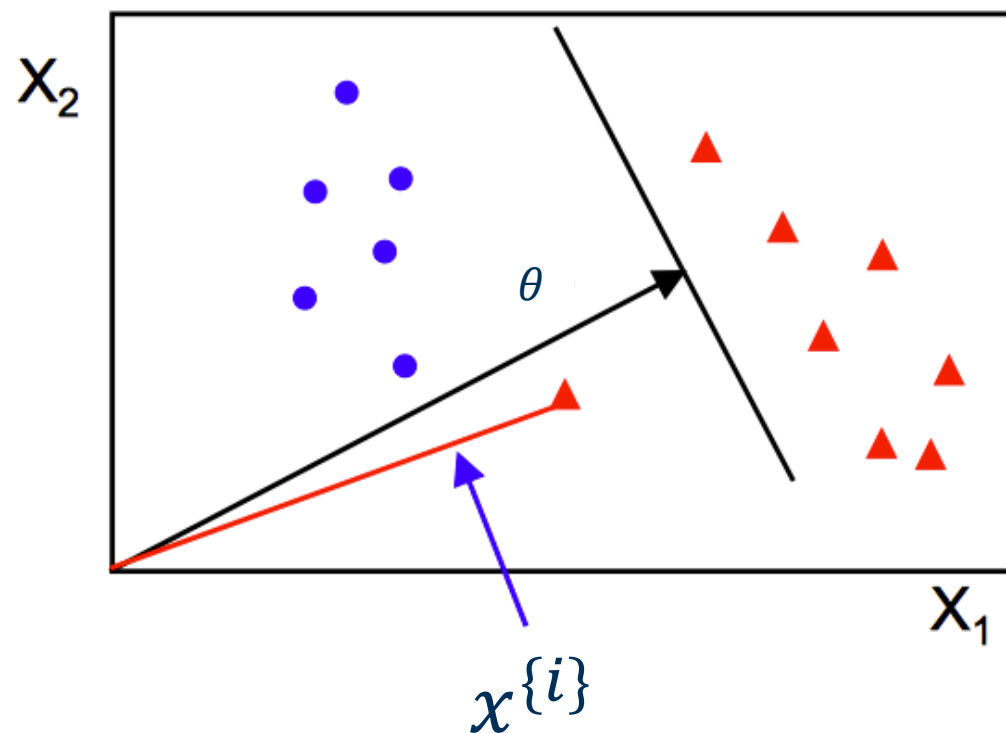
Misclassified

Ex. $y^{(i)} f(x^{(i)}) < 0$

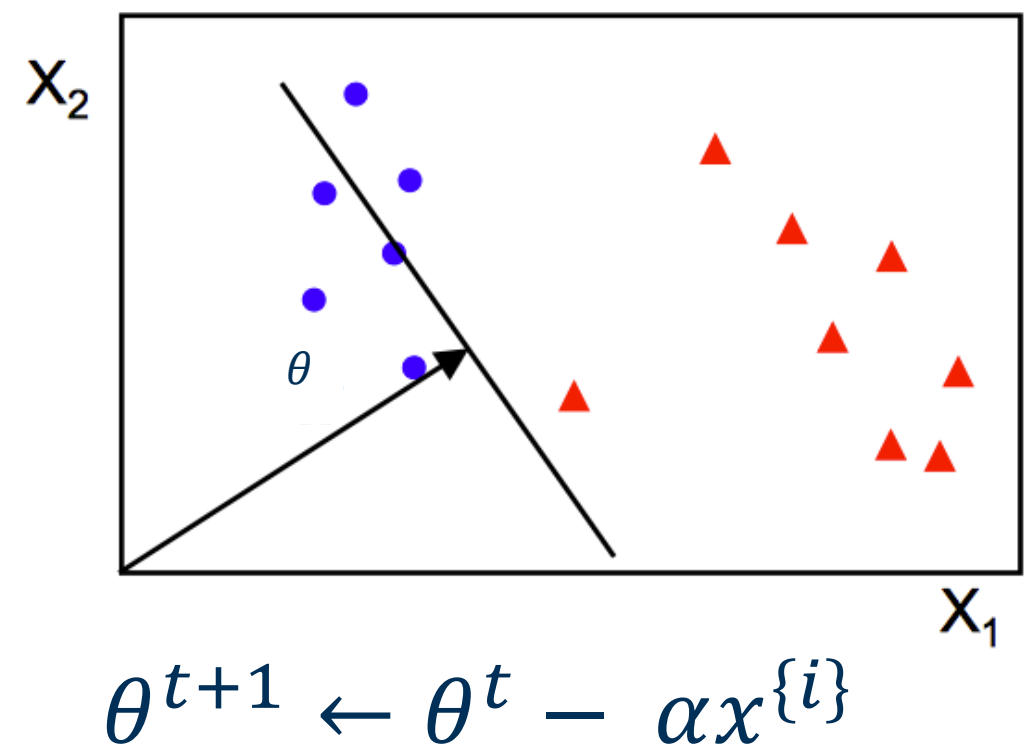
actual predicted

- Initialize θ
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
- If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified

before update



after update





**colleague
congratulating me
for achieving
state-of-the-art**

**me who trained
on test data**

Recap

- Perceptron Algo

Perceptron update

If $x^{\{i\}}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{\{i\}} x^{\{i\}}$

Misclassified point pulls θ in the direction that will flip the sign of $\theta \cdot x_i$ next time.

1. Multiply by y_i :choose the correct direction

- $y_i = +1$:move θ **toward** x_i — increase $\theta \cdot x_i$)
- $y_i = -1$:move θ **away** from x_i — decrease $\theta \cdot x_i$)

This flips the sign automatically so the dot product moves in the correct direction.

2. Multiply by x_i :fix the boundary using the misclassified point

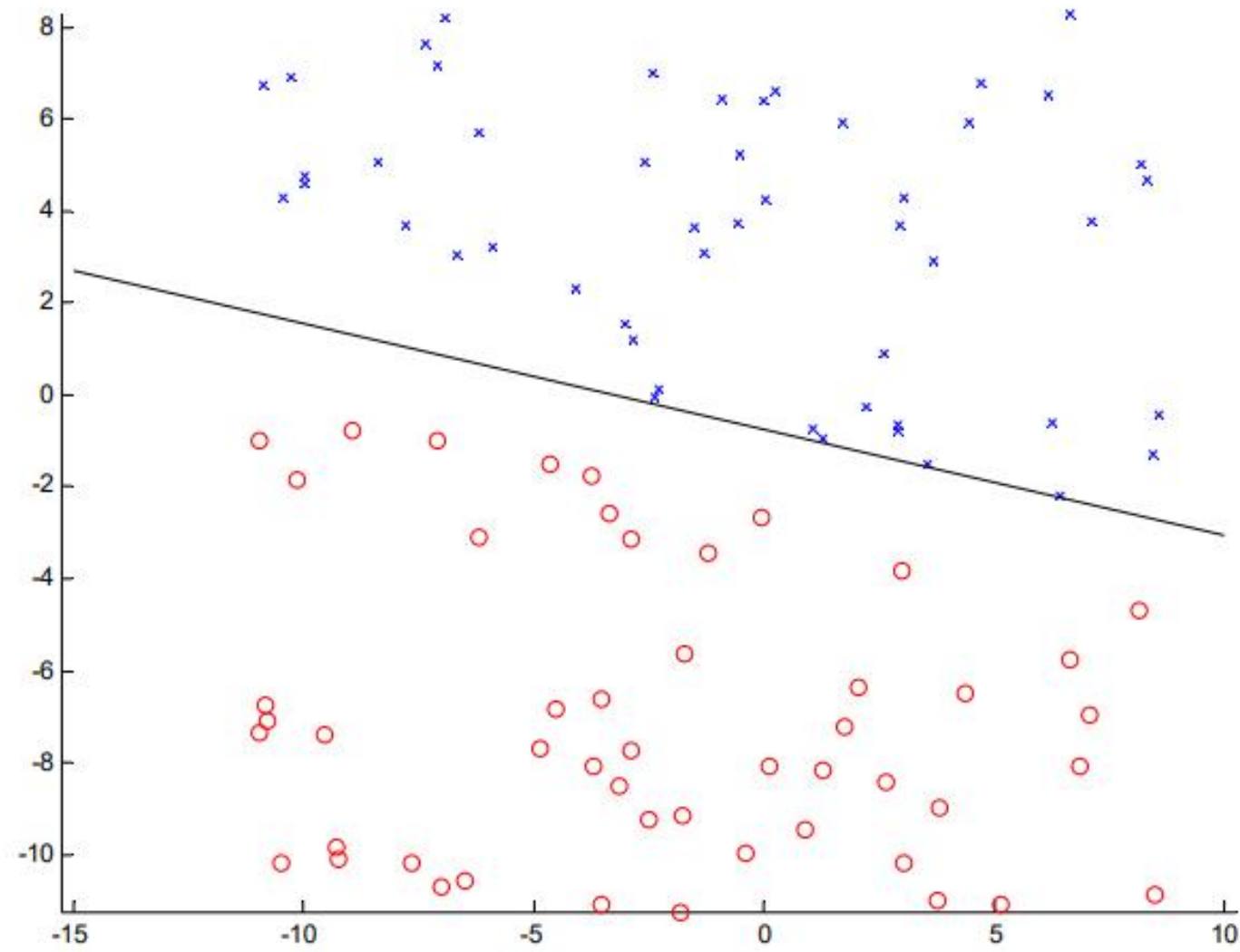
- x_i points directly to *where* the mistake occurred
- Adding $+\alpha x_i$ rotates the decision boundary toward that point
- Adding $-\alpha x_i$ rotates it away from that point

The model adjusts based on the geometry of that specific data point.

3. Multiply by α : control the step size

- Small $\alpha \rightarrow$ gentle correction. Large $\alpha \rightarrow$ aggressive boundary movement

Perceptron example

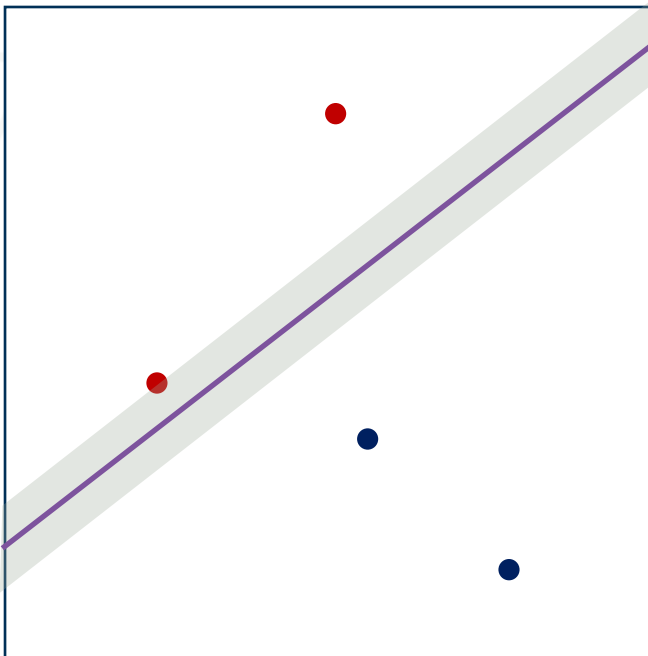


- if the data is linearly separable, then the algorithm will converge
- convergence can be slow ...
- separating line close to training data
- we would prefer a larger margin for generalization (better generalization)

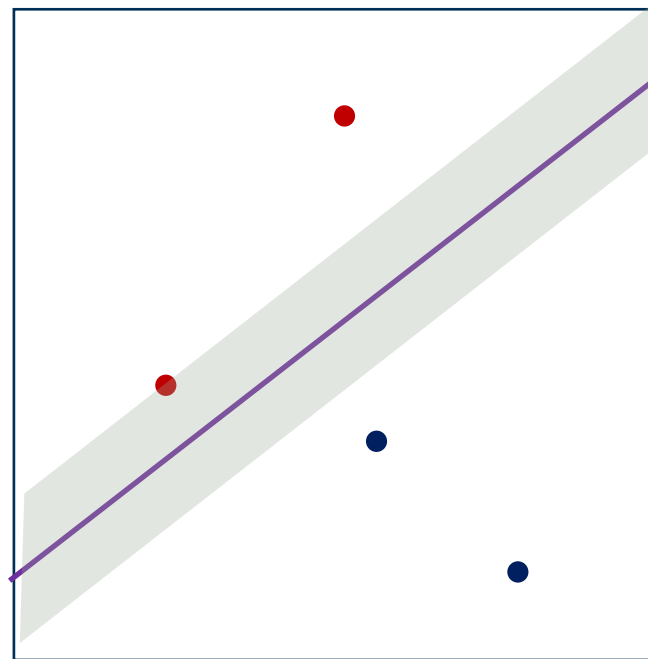
Linear separation

We can have different separating lines

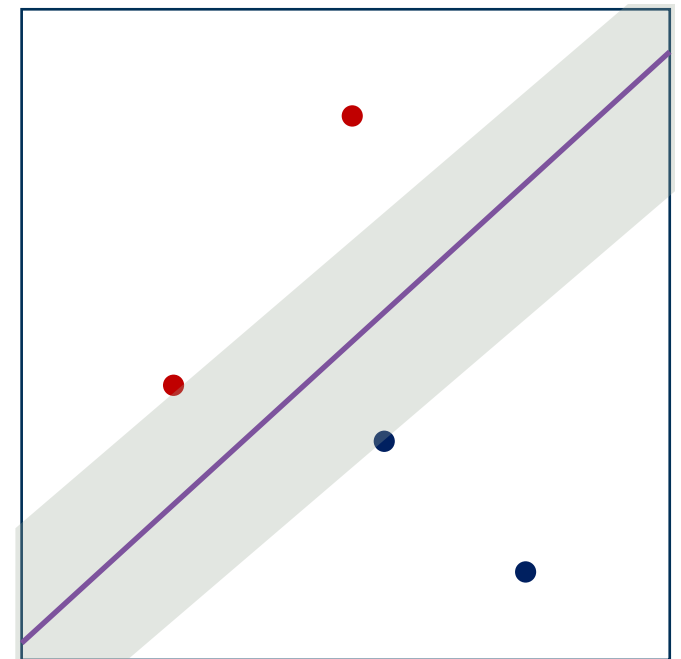
1



2



3



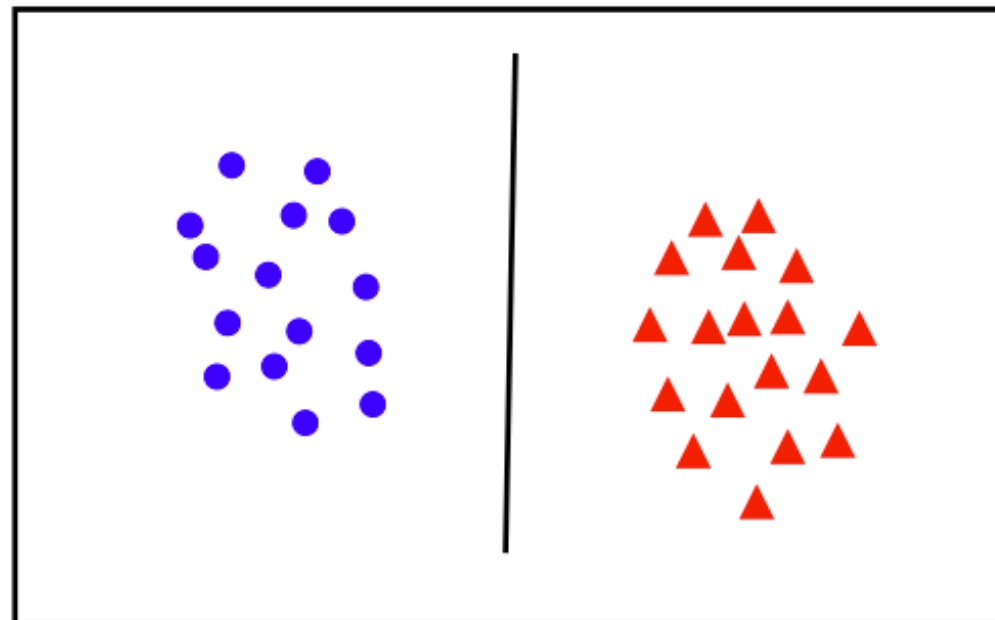
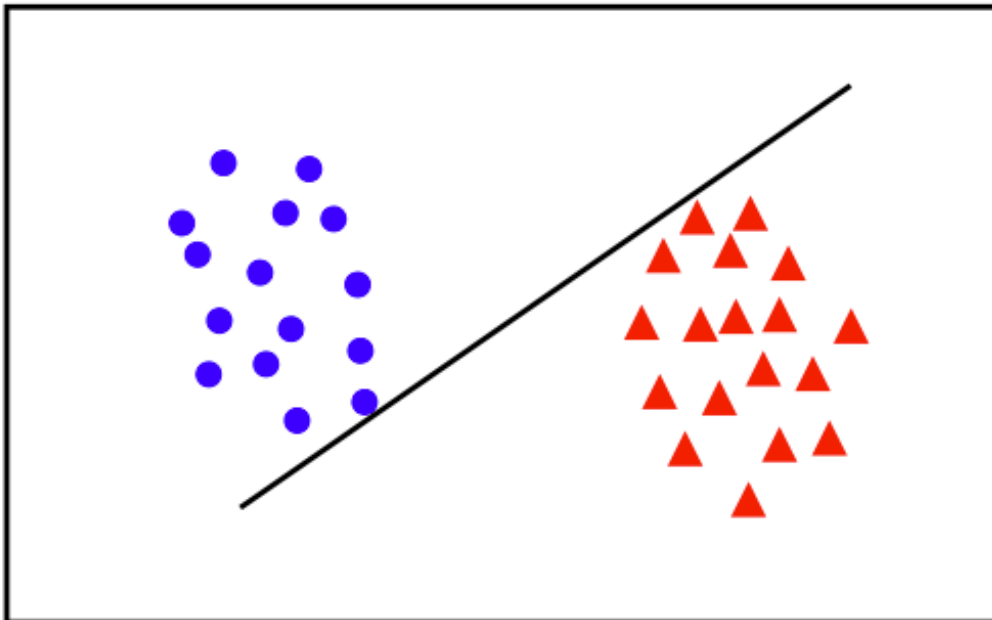
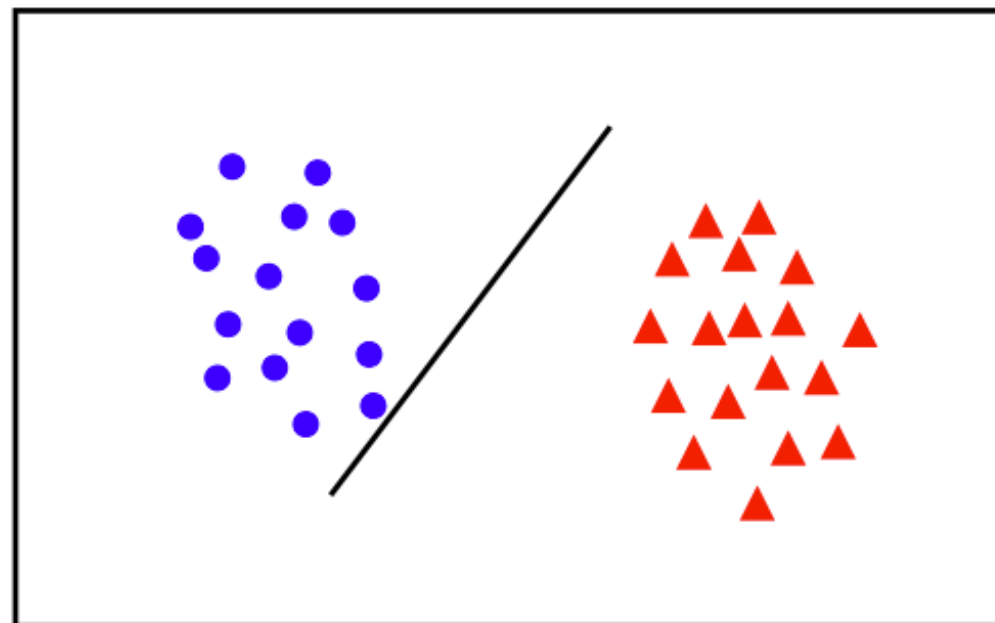
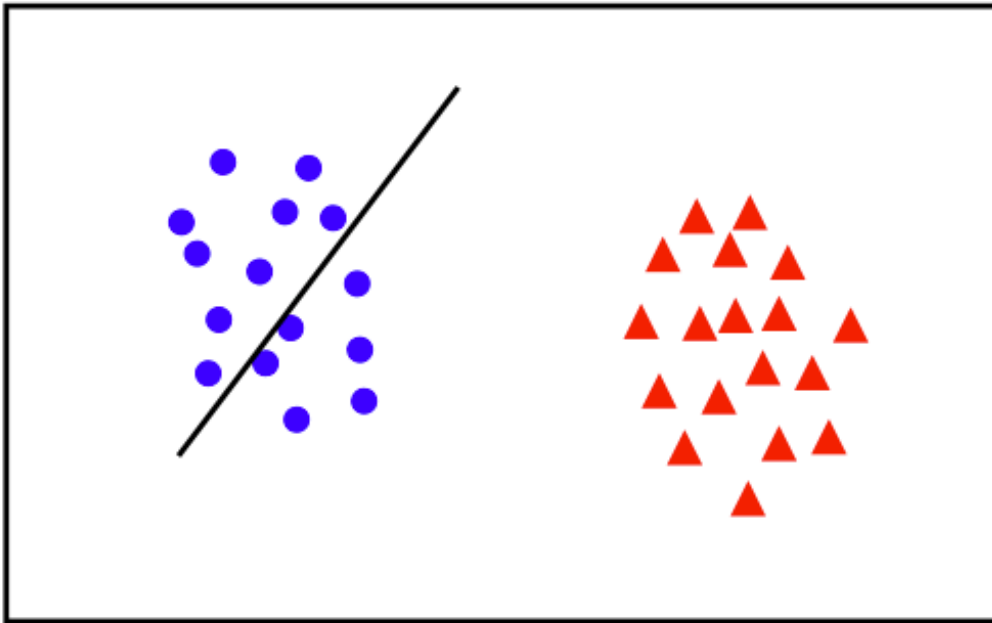
Which line is the best?

All cases, error is zero and they are linear, so they are all good for generalization.

Why is the bigger margin better?

What θ maximizes the margin?

What is the Best θ ?

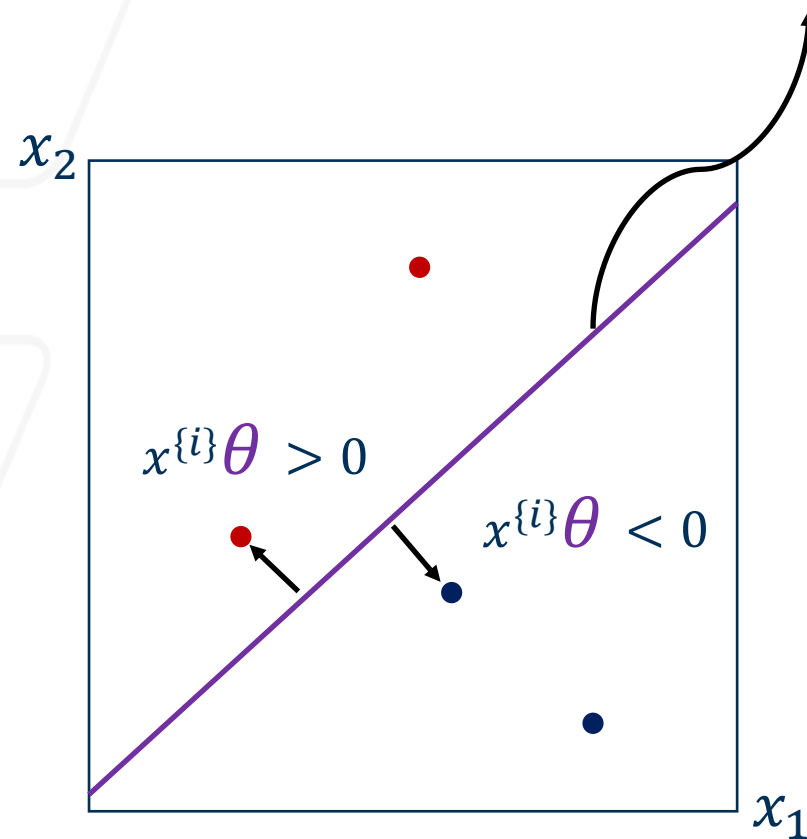


Outline

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Finding θ with a fat margin

Solution (decision boundary) of the line: $x\theta = 0$

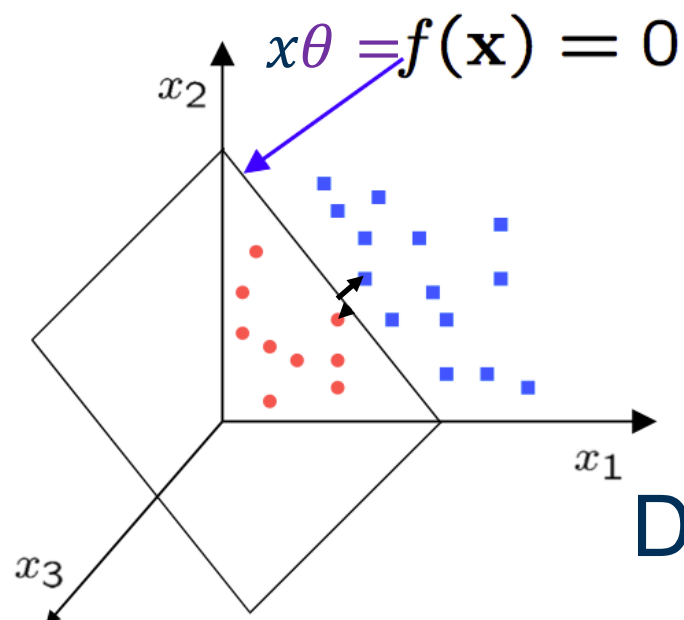


Let $x^{(i)}$ to be the nearest data point to the line (plane):

$$|x^{(i)}\theta| > 0$$

Our line solution is $x\theta = 0$

Does it matter if I scale up or down θ for the decision boundary?



$$|x^{(i)}\theta| = 1 \rightarrow \text{normalization}$$

Let's pull out θ_0 from $\theta = (\theta_1, \dots, \theta_d)$ and call it be b

Decision boundary would be: $x\theta + b = 0$

Computing the distance

The distance between $x^{\{i\}}$ and the line $x\theta + b = 0$ where $|x^{\{i\}}\theta + b| = 1$

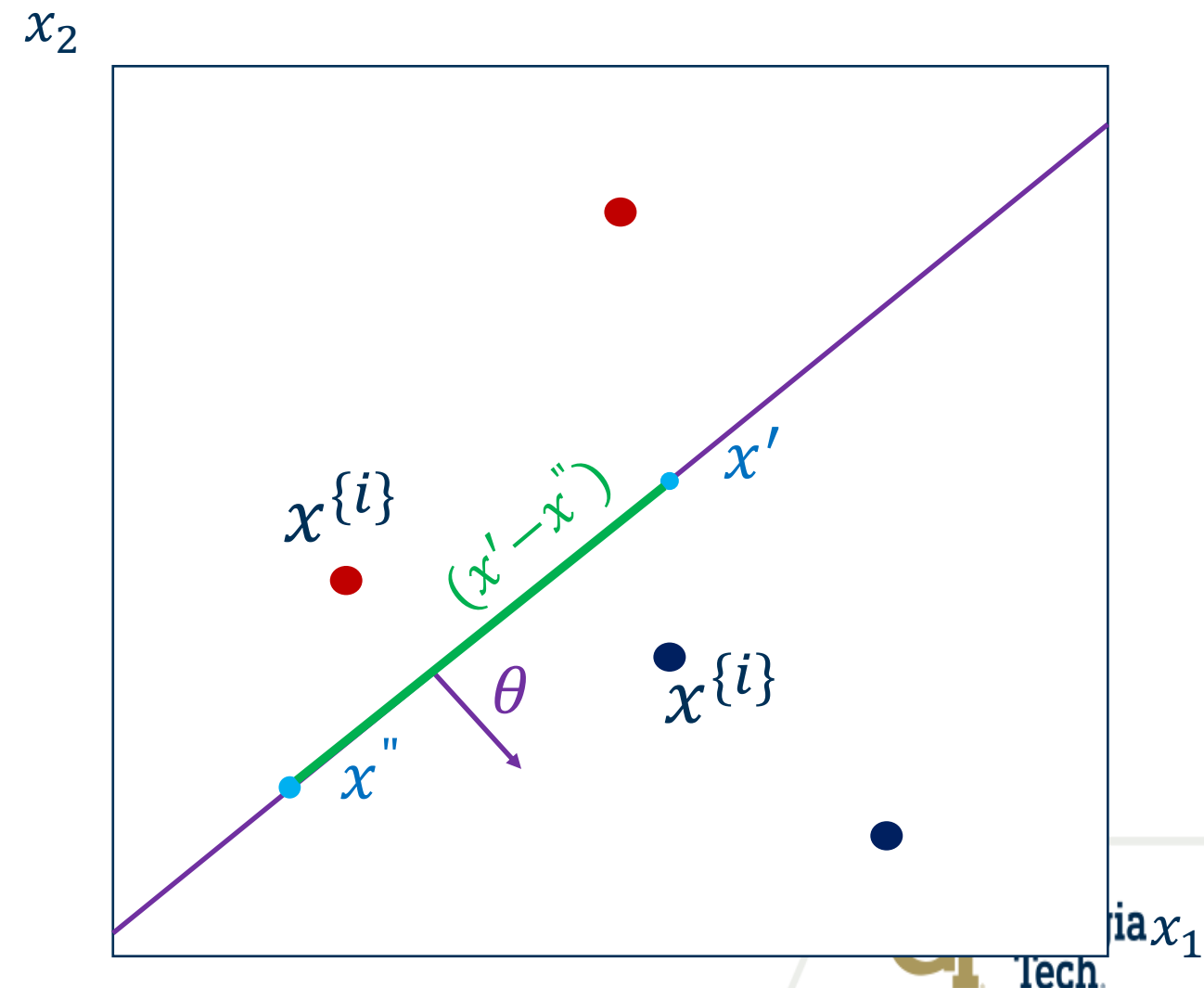
The vector θ is perpendicular to the decision line.

Consider x' and x'' on the plane

$$x'\theta + b = 0 \quad \text{and} \quad x''\theta + b = 0$$

$$\Downarrow$$
$$x'\theta + b = x''\theta + b$$

$$\Rightarrow (x' - x'')\theta = 0$$



What is the distance of my fat margin?

What is the distance between $x^{\{i\}}$ and the plane?

Let's take any point x on the line:

Distance would be projection of $(x^{\{i\}} - x)$ vector on θ .

To project the vector, we need to normalize θ to get the unit vector.

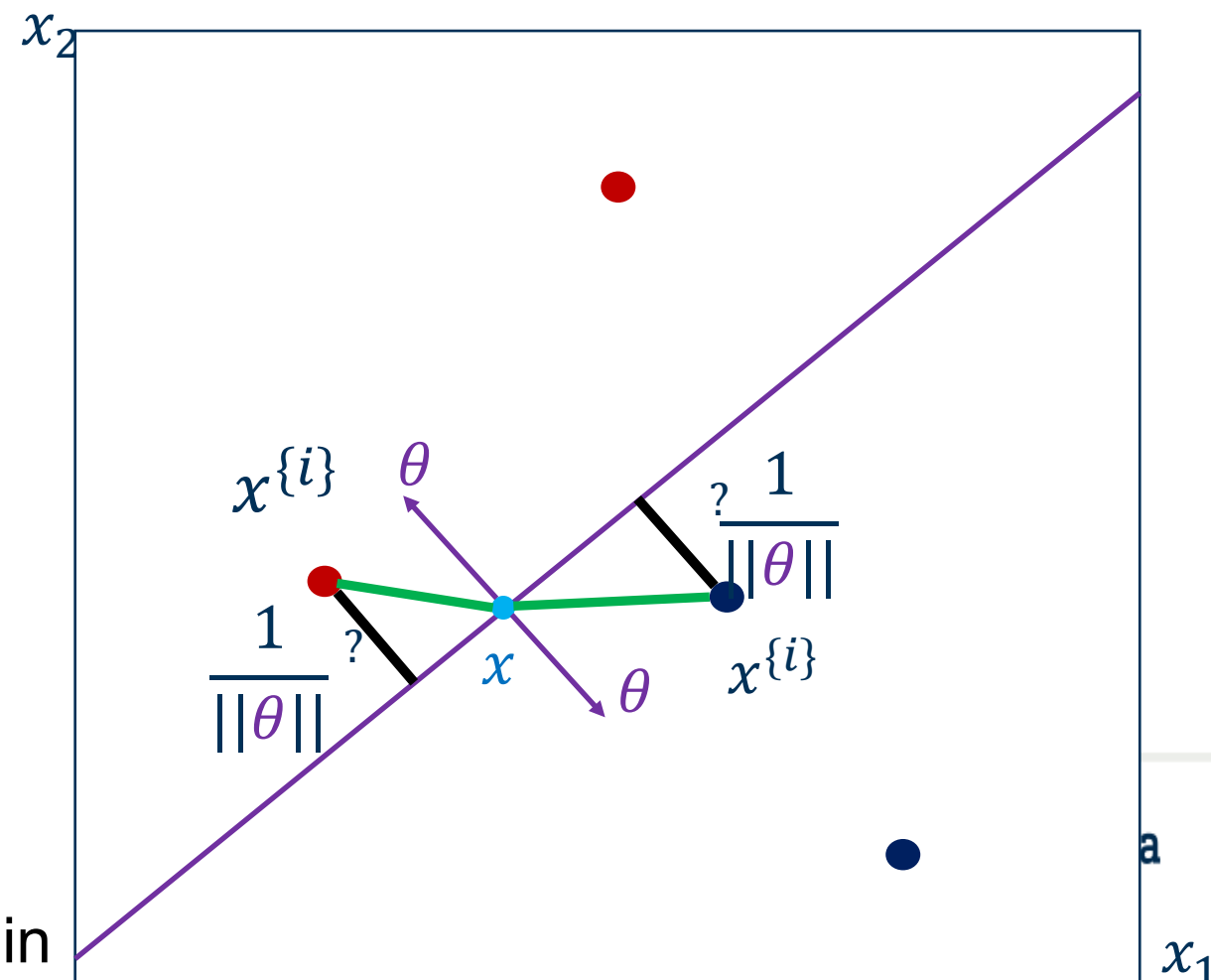
$\hat{\theta} = \frac{\theta}{\|\theta\|} \Rightarrow \text{distance} = |(x^{\{i\}} - x)\hat{\theta}|$ which is the dot product

$$\text{distance} = \frac{1}{\|\theta\|} |(x^{\{i\}}\theta - x\theta)|$$

$$= \frac{1}{\|\theta\|} | \underbrace{(x^{\{i\}}\theta + b)}_{\text{My constraint } |x^{\{i\}}\theta + b| = 1} - \underbrace{(x\theta + b)}_{\text{A point on the decision line } x\theta + b = 0} |$$

$$= \frac{1}{\|\theta\|}$$

The margin



Now we need to maximize the margin

Maximize $\frac{1}{||\theta||}$

Subject to Min value of $|x^{\{i\}}\theta + b| = 1 \Rightarrow \text{nearest neighbour}$
 $i = 1, 2, \dots, N$

There is a “min” in our constraining; it can be hard to optimize this problem(non-convex form)

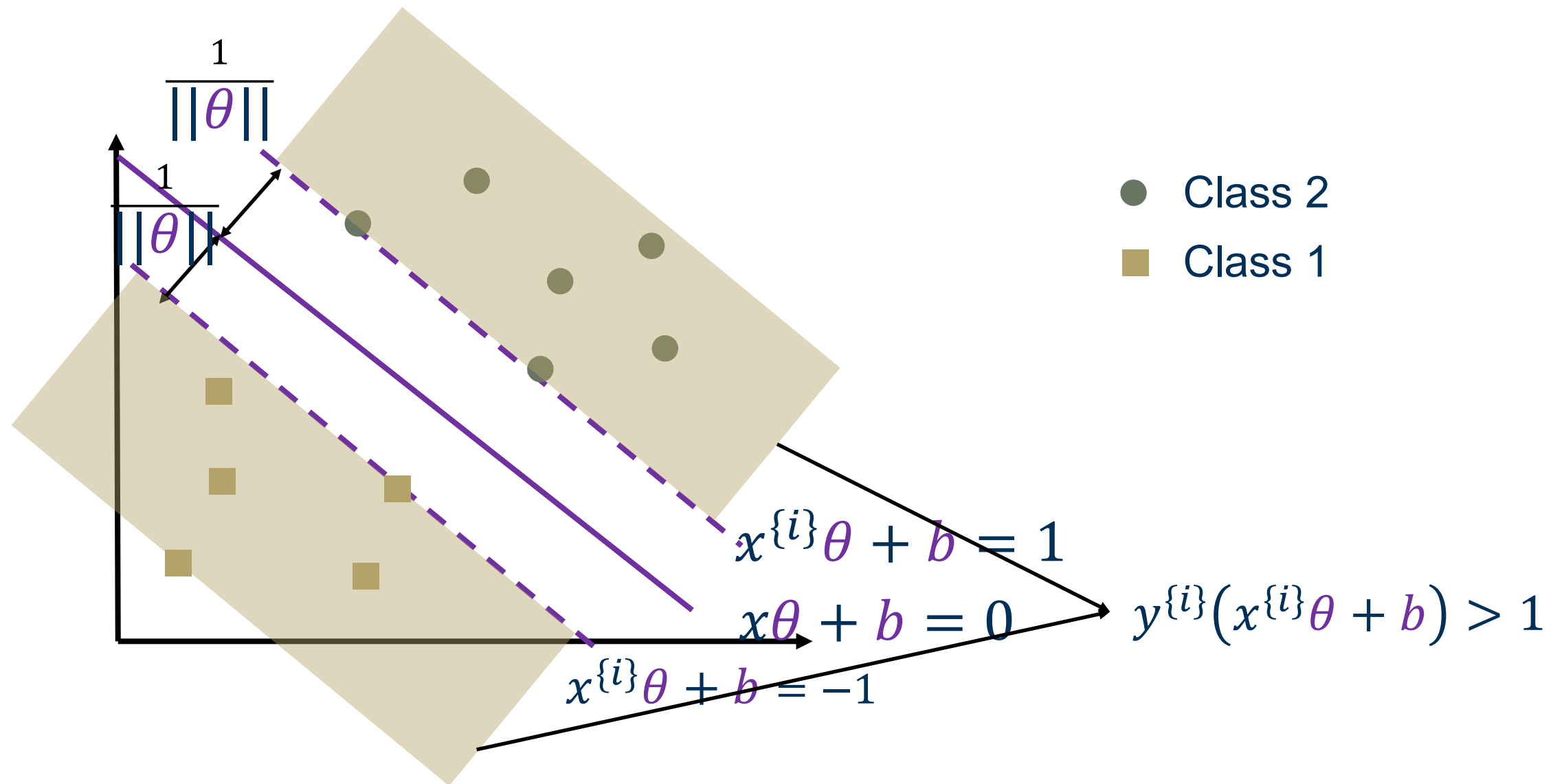
Can I write the following term to get rid of absolute value?

$$|x^{\{i\}}\theta + b| = y^{\{i\}}(x^{\{i\}}\theta + b) \Rightarrow \text{for a correct classification}$$

$$\text{If min } |x^{\{i\}}\theta + b| = 1 \Rightarrow \text{so it can be at least 1}$$

Maximize $\frac{1}{||\theta||}$

Subject to $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1 \text{ for } i = 1, 2, \dots, N$



Maximize $\frac{2}{\|\theta\|}$

Subject to $y^{i}(x^{i}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

If $\theta \neq 0$, there exists a max value

Minimize $\frac{1}{2}\theta\theta^T$

Subject to $y^{i}(x^{i}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

Constrained optimization

$$\text{Minimize } \frac{1}{2} \theta \theta^T$$

$$\text{Subject to } y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1 \text{ for } i = 1, 2, \dots, N$$

$$\theta \in \mathbb{R}^d, b \in \mathbb{R}$$

Using Lagrange method:

But wait, there is an **inequality** in our constraints

We use Karush-Kuhn-Tucker (KKT) condition to deal with this problem

$$g(x) = y^{\{i\}}(x^{\{i\}}\theta + b) - 1 \quad \alpha = \text{lagrange multiplier}$$

We need to optimize

θ, b , and α

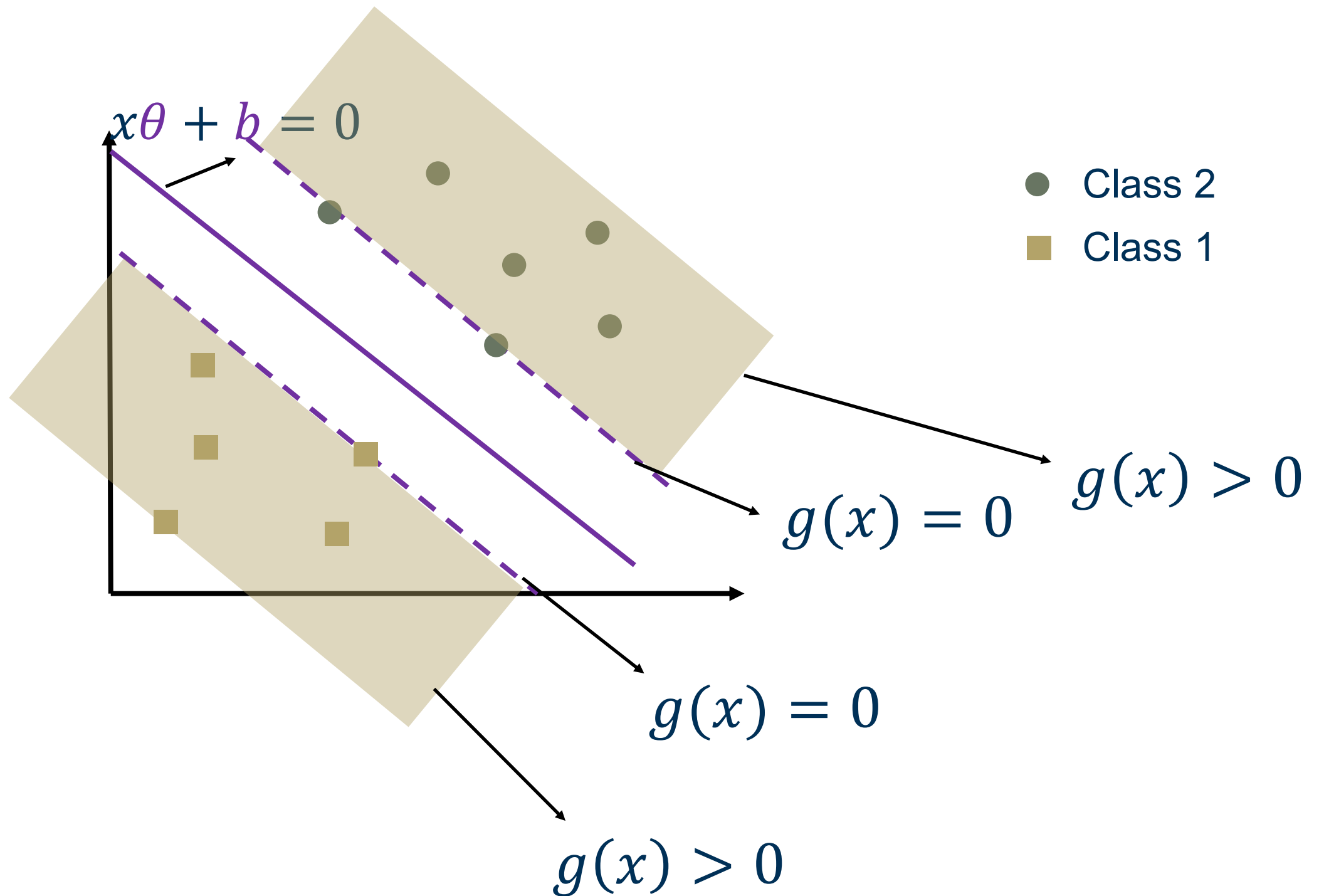
1) Stationary

2) $g(x) \geq 0$ Primal feasibility

3) $\alpha \geq 0$ Dual feasibility

4) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$



$$g(x) = y^{\{i\}}(x^{\{i\}}\theta + b) - 1$$

3) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$

Lagrange formulation

$$\text{Minimize} \quad \frac{1}{2} \theta \theta^T \quad \text{s.t.} \quad y^{\{i\}} (x^{\{i\}} \theta + b) - 1 \geq 0$$

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{\{i\}} (x^{\{i\}} \theta + b) - 1)$$

Minimize w.r.t θ and b and maximize w.r.t each $\alpha_i \geq 0$

$$\nabla_{\theta} \mathcal{L}(\theta, b, \alpha) = \theta - \sum_{i=1}^N \alpha_i y^{\{i\}} x^{\{i\}} = 0$$

$$\nabla_b \mathcal{L}(\theta, b, \alpha) = - \sum_{i=1}^N \alpha_i y^{\{i\}} = 0$$

$$\theta = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$$

$$\sum_{i=1}^N \alpha_i y^{(i)} = 0$$

Let's substitute these in the Lagrangian:

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta + b) - 1)$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta + b))$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha_i (y^{(i)} (x^{(i)} \theta))$$

$$= \sum_{i=1}^N \alpha_i + \frac{1}{2} \theta \theta^T - \theta \theta^T = \sum_{i=1}^N \alpha_i - \frac{1}{2} \theta \theta^T$$

$$\theta = \sum_{i=1}^N \alpha_i y^{\{i\}} x^{\{i\}}$$

$$\sum_{i=1}^N \alpha_i y^{\{i\}} = 0$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \theta \theta^T$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}T}$$

maximize w.r.t each $\alpha_i \geq 0$ for $i = 1, \dots, N$

and

$$\sum_{i=1}^N \alpha_i y^{\{i\}} = 0$$

The solution – quadratic programming

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}T}$$

Quadratic programming packages usually use “min”

$$\min_{\alpha} \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}T} - \sum_{i=1}^N \alpha_i$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}T} & \dots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}T} & \dots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}T} \end{bmatrix} \alpha + (-I^T) \alpha$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}T} & \dots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}T} & \dots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}T} \end{bmatrix} \underbrace{\alpha + (-I^T) \alpha}_{\text{Linear term}}$$

Quadratic coefficients

Subject to

$$\sum_{i=1}^N \underbrace{\alpha_i y^{\{i\}}}_{\text{Linear equality constraint}} = y^T \alpha = 0$$

Linear equality constraint

Pass these to a quadratic programming package

$$\text{lower bound}(0) \leq \alpha \leq \text{upper bound}(\infty)$$

$$\min_{\alpha} \frac{1}{2} \alpha^T Q \alpha - 1^T \alpha \quad \text{subject to} \quad y^T \alpha = 0; \alpha \geq 0$$

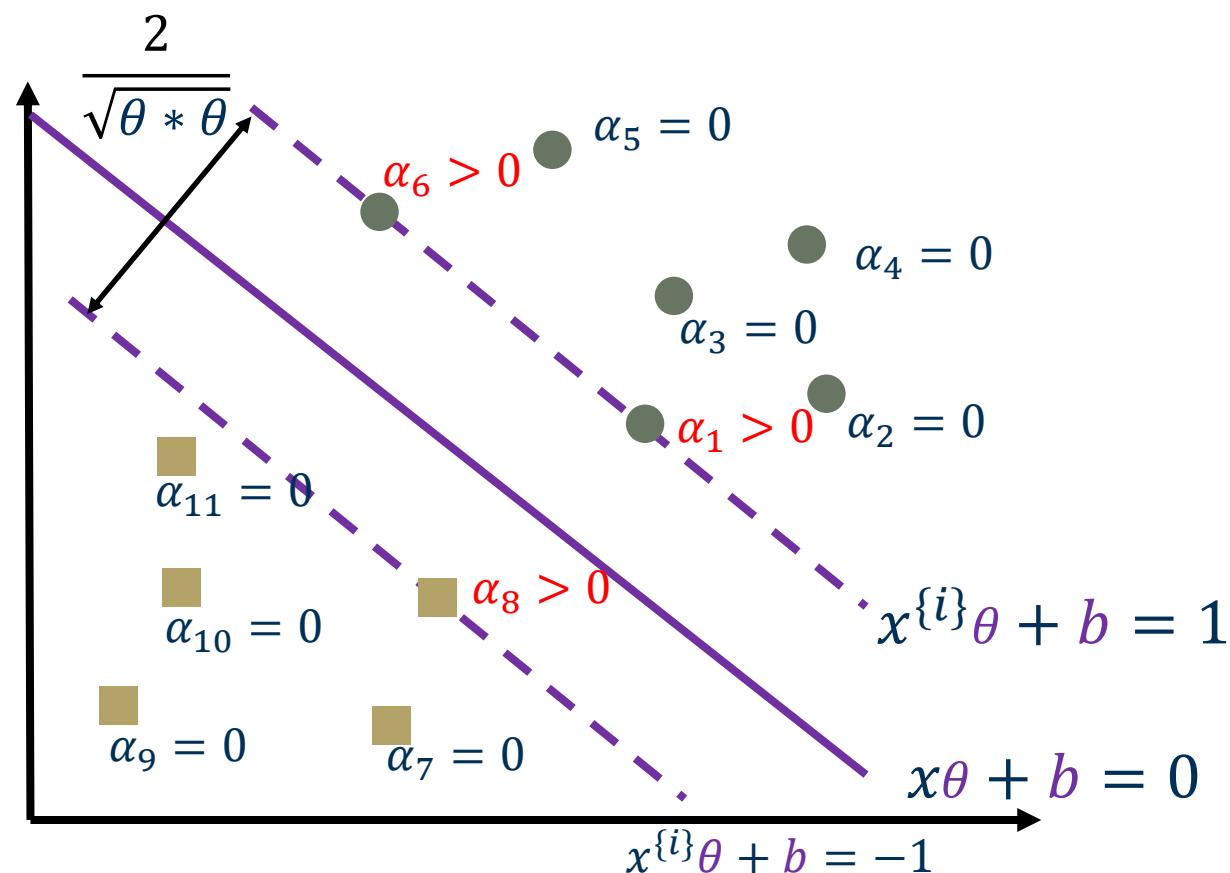
Quadratic programming will give us α

Solution: $\alpha = \alpha_1, \dots, \alpha_N$

KKT condition ($\alpha_i g_i(\theta) = 0$): $\alpha_i (y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) > 0 \quad \Rightarrow \quad \alpha_i = 0$$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0 \quad \Rightarrow \quad \alpha_i > 0 \Rightarrow x_i \text{ is a support vector}$$



Training

$$\theta = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$$

No need to go over all datapoints

$$\rightarrow \theta = \sum_{x^{(i)} \text{ in } SV} \alpha_i y^{(i)} x^{(i)}$$

and for b pick any support vector and calculate:

$$y^{(i)}(x^{(i)}\theta + b) = 1$$

Testing

For a new test point s

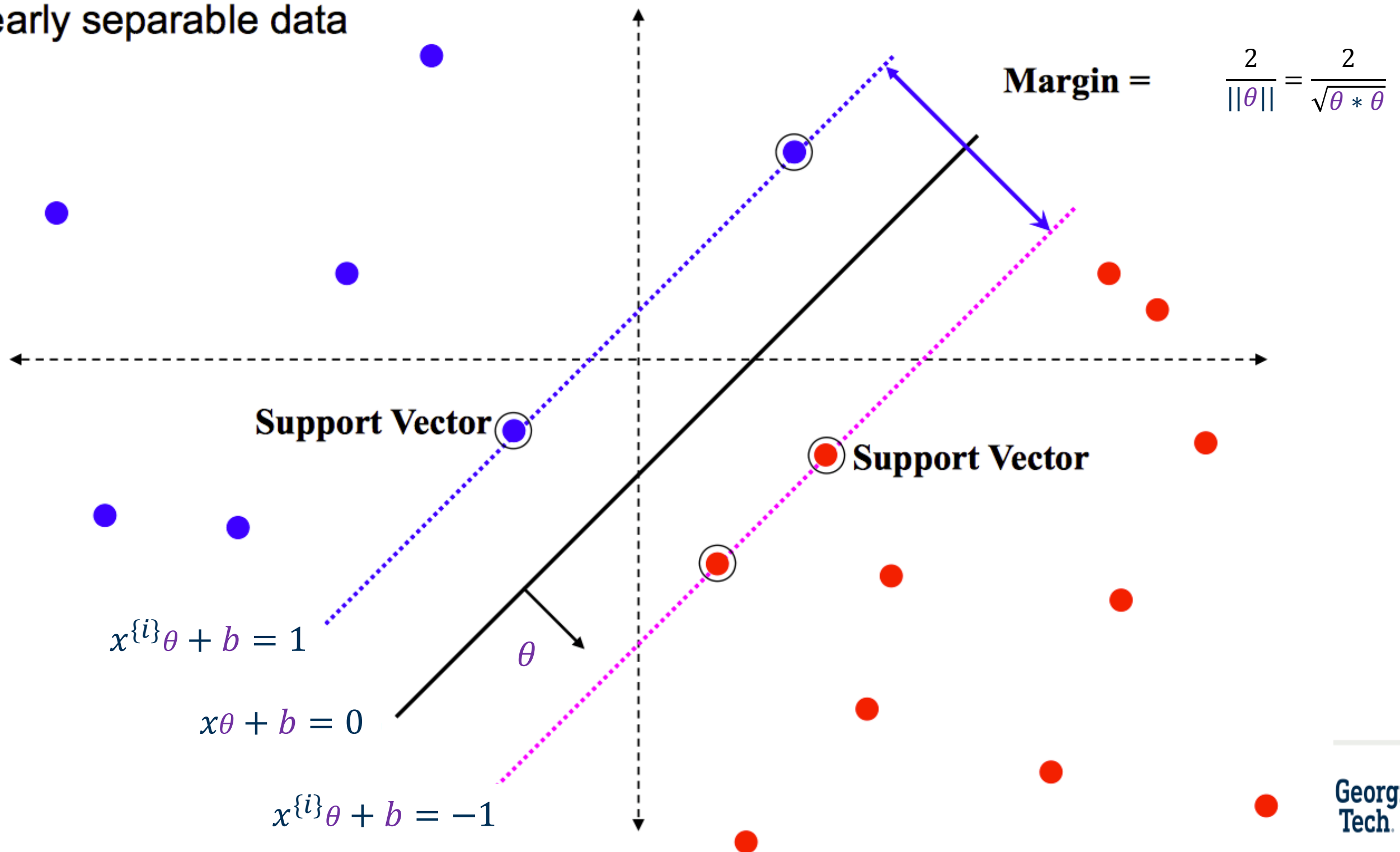
Compute:

$$s\theta + b = \sum_{x^{(i)} \text{ in } SV} \alpha_i y^{(i)} x^{(i)} s^T + b$$

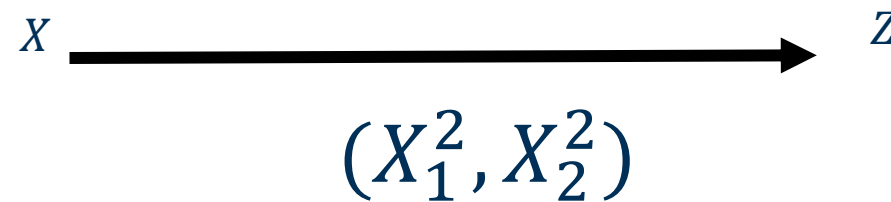
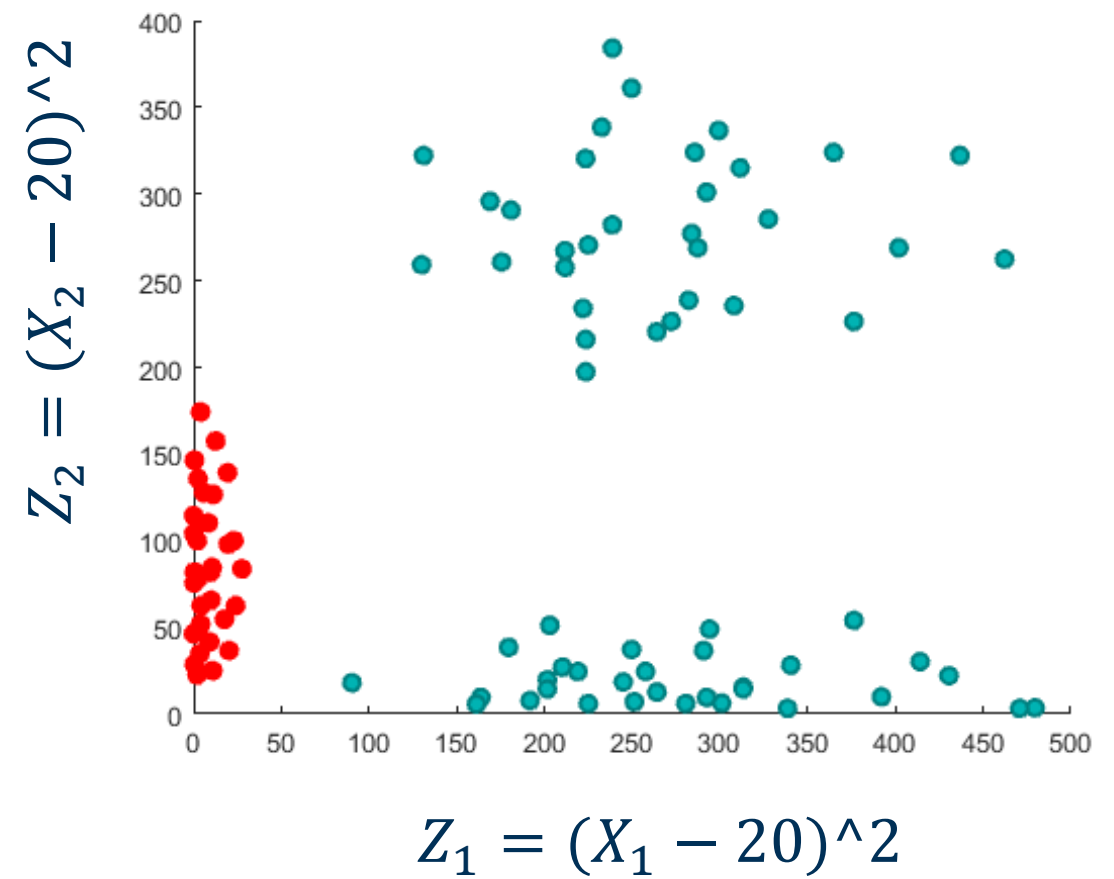
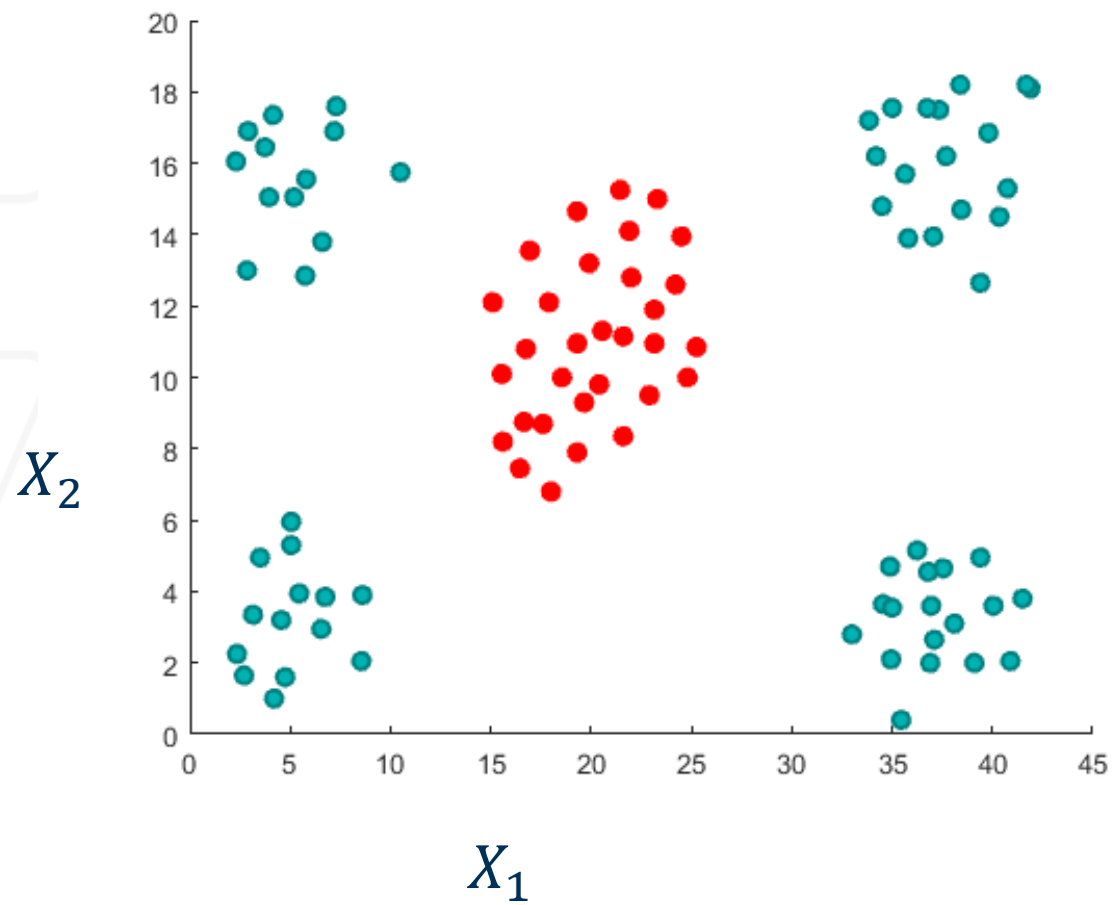
Classify s as class 1 if the result is positive, and class 2 otherwise

Geometric Interpretation

linearly separable data



From x to z space



In x space

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j x^{(i)} x^{(j)T}$$

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j x^{(i)} x^{(j)T}$$

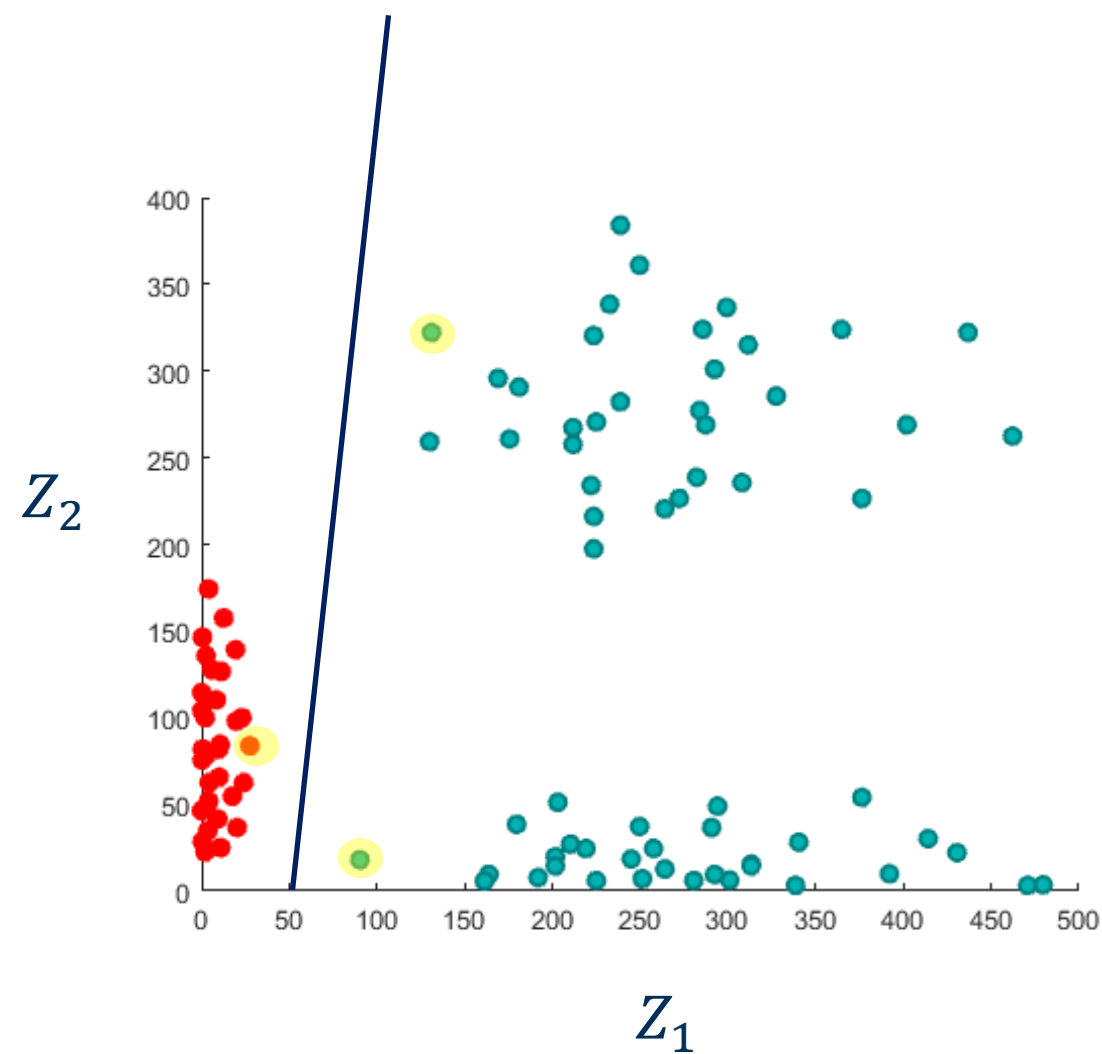
X_2 X_1

let's say x is $n \times d$
 xx^T will be $n \times n$

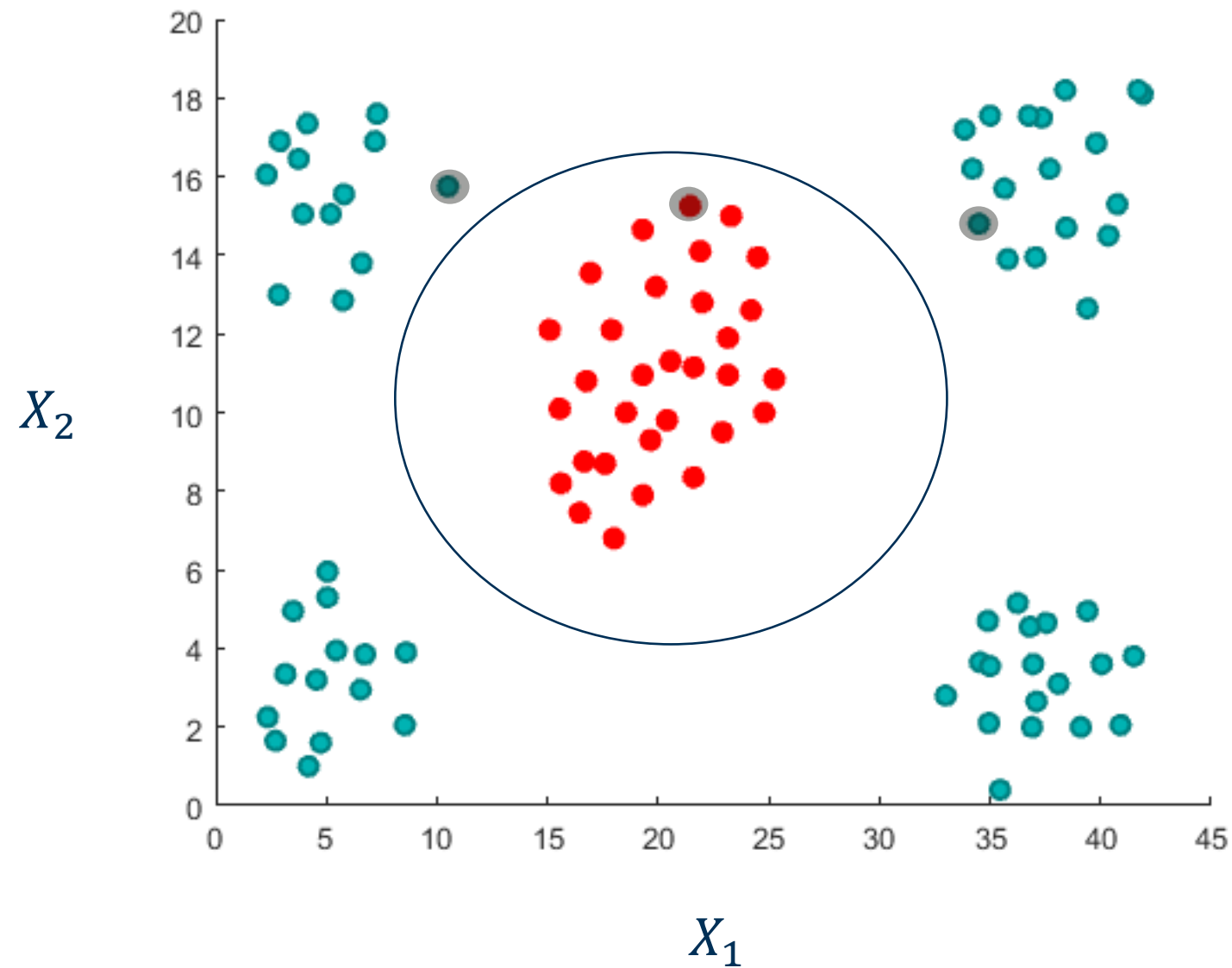
If I add millions of
 dimensions to x ,
 would it affect the
 final size of xx^T ?

In z space

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha_i \alpha_j z^{(i)} z^{(j)T}$$



In x space, they are called pre-images of support vectors



Take-Home Messages

- Linear Separability
- Perceptron
- SVM: Geometric Intuition and Formulation

Quick Knowledge Check

**Please fill out CLOS. Potential Extra Credit Opportunity
will be released on Ed**

Remaining things in the semester

- Kernel SVM will be released. **Please go through it.**
 - How to use kernel trick to use SVM for non linearly separable data
 - Take data from x space to z space (higher dimension)
 - High chances of data to be linearly separable in z space
 - Do SVM in that space
 - Use kernel trick
- Quiz 14 – This weekend
- Final Presentation with your mentor – Link on Ed
- Final project – Due Dec 2nd
- Peer Evaluation – Due Dec 2nd

Thank you for a wonderful semester!😊

Good Luck in completing your projects and for all the success you are about to achieve!

